

8 (a) energy required to separate nucleons in a <u>nucleus</u> to infinity	M1 A1	[2]
(b) $1\text{u} = 1.66 \times 10^{-27} \text{ kg}$ $E = mc^2$ $= 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$ $= 1.49 \times 10^{-10} \text{ J}$ $= (1.49 \times 10^{-10}) / (1.6 \times 10^{-13})$ $= 930 \text{ MeV}$	C1 M1 M1 A0	[3]
(c) (i) $\Delta m = 2.0141\text{u} - (1.0073 + 1.0087)\text{u}$ $= -1.9 \times 10^{-3} \text{ u}$ binding energy $= 1.9 \times 10^{-3} \times 930$ $= 1.8 \text{ MeV}$	C1 A1	[2]
(ii) $\Delta m = (57 \times 1.0087\text{u}) + (40 \times 1.0073\text{u}) - 97.0980\text{u}$ $= (-)0.69\text{u}$ binding energy per nucleon $= (0.69 \times 930) / 97$ $= 6.61 \text{ MeV}$	C1 C1 A1	[3]

Page 5	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2011	9702	42

Section B

0 (a) thin / fine metal wire

B1